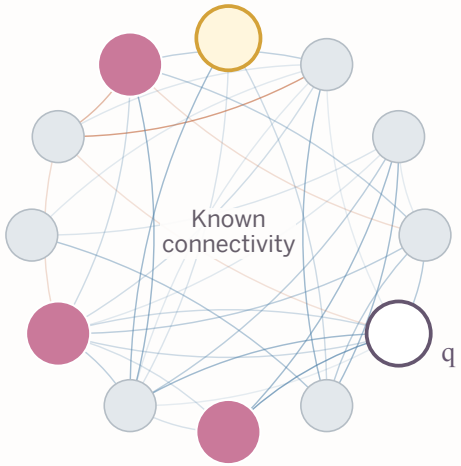
Predictive risk and recording decisions

in partially observed neural networks



100 simulated units · M = 5

Uncertain neuronal gains

- Recorded node
- Stimulated node
- Response readout (one of three)

1 Endpoint recording

Stimulation site and readouts

Lower held-out error

98% 37 of 40 networks improved



Both endpoints vs neither, mean MSE

`-98.13% [-99.19, -93.94]`

Narrower feasible range

99% 40 of 40 networks narrowed



Mean squared width, both vs neither

`-99.08% [-99.62, -96.10]`

Network not improved

2 Error factors

Endpoint vs ordinary recording
Exact three-factor split

Input factor

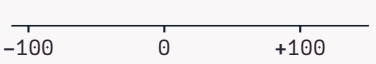
`-98.7%  [-99.5, -96.3]`

Readout factor

`-98.8%  [-99.6, -96.9]`

Recurrent factor

`+16.5%  [-34.6, +139.4]`



Mean-square change vs ordinary (%)

95% paired network-bootstrap CI

3 Fifth recording

Endpoints kept, one site added
80 new independent networks

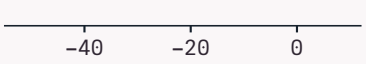
63 of 80 networks improved combination estimates



Fifth site chosen for recurrent gain sums, compared with a random fifth site

Combination estimation MSE

`-31.1%  [-42.4, -19.8]`



Change vs random fifth site (%)

95% paired network-bootstrap CI

Integrated stimulus response

$$q = \int \Delta r(t)\, dt$$

$$\Delta r(t) = r_{\text{stim}}(t) - r_{\text{control}}(t)$$

Better risk estimates do not necessarily make better recording decisions.

We used a **common MAP estimator** in 12-unit networks, so identical recordings and data yield identical predictions. The two criteria chose identical recordings in **68% and 65%** of designs, with no clear change in error (−3.7%, +2.7%).

Recording the stimulation site and readouts lowered held-out response error by **98%** in independent networks. A fifth site chosen for recurrent gain sums improved their estimates by **31%**, with no clear change in response error.